Lecture 12: Evolution, Emotion, and Reason: Emotions, Part II (full transcript)

Source: https://oyc.yale.edu/psychology/psyc-110/lecture-12
License: Creative Commons Attribution-NonCommercial-ShareAlike 3.0 (Open Yale Courses)

Overview: Professor Bloom continues the discussion of emotions as useful evolutionary adaptations for dealing with our social environment. In particular, this lecture describes evolutionary explanations for several important emotional responses, such as the love between parents and their offspring, the gratitude we feel towards cooperative behaviors, the spite we feel for cheaters, and the cultural differences in feelings of revenge.

Chapter 1. Emotional Responses to Caregivers and Kin [00:00:00]

Professor Paul Bloom: So what we're doing today is continuing on the theme of emotions. "Emotions" is a two-part lecture and we're continuing along certain themes.

I want to begin by responding to a question which was raised in the last class concerning smiling and nonhuman primates. It was a very good question. The issue was: we know that humans have different sorts of smiles to convey different sorts of information. The question was, "do nonhuman primates, like chimpanzees or gorillas or gibbons, have the same many sorts of smiles?" So, I contacted the world's expert on smiling, who did not return my e-mails. So, I contacted the second world's expert on smiling who told me that the answer is "no," that primate — nonhuman primate smiles actually correspond almost entirely to appeasement smiles. They're "don't hurt me" smiles. They're equivalent to the "coy smile" that we saw on humans. But that nonhuman primates do not use smiles for greetings; there's no equivalent to the "greeting smile" or "Pan Am smile"; nor do they use them as genuine expressions of happiness. There's no equivalent to the "Duchenne smile." That's as far as I know. If the world's expert gets back to me and says something different, I'll keep you posted.

Another thing. Going back to the beginning theme of the class, what we started — just to review, we talked about the different functions of emotions. And then we talked about smiling and facial expressions. And then we turned to some — to a nonsocial emotion, the case of fear. And then we shifted to social emotions. And we talked about social emotions towards kin and the special evolutionary reasons that would lead them to evolve. And as we were ending, we were talking about the relationship between an animal and its children, particularly in cases like humans and birds and mammals where there tends to be a close relationship with our children.

We invest in quality, not quantity. I might produce very few children in my life. And my evolutionary trick then is to focus very intently on them and make sure they survive. If I were to produce 100 children, I could stand to lose a few, but if I just produce five in my lifetime or two or one, they become very precious to me. And so, the story of the evolution of a species like us involves a long period of dependence and deep, deep bonds between the parent and the child. And that's part of what I talked about, how parents respond to children.

And I want to begin this class by giving an illustration from a documentary about parental response to children, but I want to give it in a species that's not us. And here is why. I'll explain why with an analogy. I have a friend of mine who studies the psychology of religion. He studies why people hold

religious beliefs. And he tells me that when he's talking to a non specialist, somebody not in the field, he doesn't ever tell them, "Yeah, I'm really interested in why people believe in the Bible or why people light the candles on Sabbath or why people go to church" because these are religions that people around here hold, and if you tell people you study them they'll sort of be puzzled, "why would you want to study something like that" or offended. If you want to talk about the psychology of religion to an audience like this, what you do is you start with the exotic. So, you start by talking about people who put butter on their heads.

Dan Sperber talks about a culture where the men put butter on their heads in the summer. And it kind of melts and that's part of — one of the things that they do or — you talk about a culture that believes in spirits or that trees can talk. You say you're studying it and they say, "Oh, that's interesting. I wonder why they believe that?" And you use that as a way to look at more general facts that exist even in our culture. You use the fact that we don't take the exotic for granted as a way to motivate the scientific study of things we do take for granted.

And this is, of course, true more generally. This was the point in the William James quote when he talked about things that are natural to us and noticed that some very odd things are equally natural to other species. And it's true, I think, in particular when we talk about things like the love we have for our children. So, one way to look at the love we have for our children scientifically, isn't to look at it head-on, because the love we feel towards our own children feels sacred, it feels special, but look at it in other species. And so, one of the nicest illustrations of this is the Emperor penguin, which was — which — whose childcare and mating practices were dramatized in a wonderful movie called "March of the Penguins." And this is interesting because they had this incredibly elaborate and quite precarious system of generating and taking care of offspring.

So, I want to show you a brief clip of the movie to illustrate some parts of this. What they do at the beginning, which is not — which leads up to this, is they take a very long trek from the water to their breeding grounds. Their breeding grounds is — are protected from the wind and they're on a firm piece of ice so they could hold the whole pack. They do the breeding there and it's there that the eggs are created. So, this is where the movie begins at this point. [clip playing]

"March of the Penguins" was the second best — second most popular documentary of all time, beaten only by "Fahrenheit 9/11." And people responded to it in different ways, which are informative when we think about the generalizations you could make from animal behavior to human behavior. Some conservative commentators saw this as a celebration of family values, such as love and trust and monogamy. Some liberals, who hate everything that's good and true, [laughter] responded by saying, "Well, yeah, they're monogamous for one breeding season. It's a year. Then they go and find another mate. If you add it up, it's pretty slutty." [laughter]

I think more to the point, people were impressed and stunned by the rich and articulate and systematic behavior that these animals were showing. Plainly, they didn't pick it up from television, movies, culture, learning, schooling, and so on. To some extent, this sort of complicated behavior came natural to them. And it's understandable that some proponents of intelligent design, or creationism, pointed to this as an example of how God creates things that are deeply, richly intricate so as to perpetrate the survival of different animals. From a Darwinian standpoint, the Darwinian would agree with the creationist that this couldn't have happened by accident, this is just far too complicated, but would

appeal to the — to this as an exquisite example of a biological adaptation, in particular a biological adaptation regarding parental care to children shaped by the fact that children share the parents' genes and so parents will evolve in ways that perpetrate the survival of their children.

Then there's the other direction, which is how children respond to parents, how the young ones are wired up to resonate and respond in different ways to the adults around them. And we quickly talked about some different theories of this. And I'll just review what we talked about last class. Babies will develop an attachment to whoever is closest. They'll usually prefer their mothers because their mothers are typically those who are closest to them. They'll prefer her voice, her face, her smell. It used to be thought that there is some sort of magical moment of imprinting that when the baby is born, the baby must see his or her mother and "boom," a connection is made. If the baby doesn't, terrible things will happen with attachment later on. This is silly. There is no reason to believe there's some special moment or special five minutes or special hour. It's just in the fullness of time babies will develop an attachment to the animal that's closest to it. They will recognize it as, at an implicit level, at an unconscious level, as their kin.

Well, how does this work? How does the baby's brain develop — come to develop an emotional attachment to that creature? Well, you remember from Skinner that operant conditioning could provide a good answer to this. And this is known as the "Cupboard Theory," which is babies love their moms because their moms provide food. It's the law of effect. It's operant conditioning. They will approach their mothers to get the food from them. And they will develop an attachment because their mother provides food. And this is contrasted with a more nativist, hard-wired theory developed by Bowlby which claims that there's two things going on. There is a draw to mom for comfort and social interaction and fear of strangers.

Now, in the real world, it's difficult to pull apart these two means of attraction because the very same woman who's giving you comfort and social interaction is also the one giving you milk. But in the laboratory you can pull them apart. And that's what Henry Harlow did in the movies you saw last week. So, Harlow exposed primates to two different mothers. One is a wire mother. That's a Skinnerian mother. That's a mother who gave food. The other is a cloth mother set-up so that she'd be comfortable and give warmth and cuddling. And the question is, "Which one do babies go for?" And as you can remember from the movies, the results are fairly decisive. Babies go to the wire mother to eat — as one of the characters said, "You've got to eat to live." But they viewed the — they loved the cloth mother. They developed an attachment to the warm, cuddly mother. That's the one they used as a base when they were threatened. That's the one they used as a base from which to explore.

Okay. And that actually — Oh, that's just — I have a picture. And that actually takes me to the — Oh, except for one thing, it almost takes me to the end of the question of our emotions towards kin. One question you could ask is, "What if there's no contact at all?" Now, you could imagine the effects of how — A lot of people are interested in the question of the effects of the child's early relationship to adults around him or her in how the child turns out later. This becomes hugely relevant for social debates like daycare. So for instance, a lot of psychologists are interested in the question, "Is it better for a child to be raised by a parent, usually a mother, or does it make a difference if the child goes to daycare? What if the child goes to daycare at six months? What if the child goes to daycare at two years? How does this affect the child?" The short answer is, nobody really knows. There's a lot of debate over whether or not

there are subtle differences and it's deeply controversial. But we do know that it doesn't make a big difference. We do know that if you got raised by mom, or perhaps mom and dad, or maybe just dad all through your life until going off for school and I — my parents threw me in a daycare at age three months — it's not going to make a big difference for us, maybe a subtle difference though it's not clear which way it would go. But it won't make a big difference.

But what if there's no contact at all? What if — What about terrible circumstances where people get no cloth mother, they get nobody for attachment? This is a really — In the real world, of course, you can't do experiments on this. And in the real world with humans, this only happens in tragic cases. But this has been studied. So Harlow, again, raised monkeys in solitary confinement so they were raised in steel cages with only a wire mother. In other words, they got all the nutrition they needed but they got no mothering. It turned out that you kind of get monkey psychotics. They're withdrawn. They don't play. They bite themselves. They're incompetent sexually. They're incompetent socially. They're incompetent maternally. In one case, one of these monkeys raised in solitary confinement was artificially inseminated. When she had a child she banged its head on the floor and then bit it to death. So, you need to be — you need — This shows — This is kind of a stark demonstration that some early connection, some early attachment is critical for the developing of a primate.

Obviously, you don't do these experiments with people but there are natural experiments, humans raised in harsh orphanages with little social contact, and these children — If the — In other words, they get fed, barely, but nobody picks them up and cuddles them. These children, if this happens for long enough, they end up with severe problems with social and emotional development. From an emotional point of view, they're often insatiable. They really need cuddling and support or they're apathetic, they don't care at all. Now, there's some sort of good news, which is if you get these people or these monkeys early enough you can reverse the effects of this bad development. So, there's some research done with monkey therapists. So then, what they do is they take the monkey, they raise it in a steel cage, the monkey comes out, the monkey is kind of psycho, and then they send in a younger monkey who is just goofing around, jumping all around the place and everything. And experience with this younger monkey who just follows them around and clings to them leads to gradual improvement. It makes the solitary monkey become better.

There might be a similar effect with humans. So one story more about — of an anecdote than an experiment was a situation where at the age of one and a half, children were taken away from a really harsh orphanage where they had no contact and brought into a home for mentally retarded women where these women gave them plenty of contact and cuddling and apparently, from what we know, brought them back to normal. And this is all I want to talk about, about the emotions we feel towards our kin, towards our children, and towards our parents.

Chapter 2. Question and Answer on Emotions towards Kin [00:15:44]

Any questions or thoughts? Yes.

Student: Do children in orphanages comfort each other?

Professor Paul Bloom: It's a good question. Do children in orphanages comfort each other? I don't know. The situation probably wouldn't be there — The problem is children in orphanages who are in these

terrible situations tend to be babies and very young and they wouldn't be thrown together in situations where they could comfort each other. It's a really interesting question. What if it was a situation where children were raised without a supportive cloth mother at all, would not be able to pick them up and hold them, but they could play amongst themselves and support each other? I don't know the answer to that.

Teaching Assistant: Yes.

Professor Paul Bloom: Yes? Is there evidence on that?

Teaching Assistant: Yes, there is. [inaudible]

Professor Paul Bloom: Yes. [laughter] The answer is there is evidence, [laughs] as everybody knows, [laughter] that this sort of — amongst the young, support can actually help the monkey and the children. Somebody else had a question here? Yes.

Student: What does that tell us about the middle ground, if the parent is comforting just a little bit and then not that much [inaudible]

Professor Paul Bloom: Right. So this is — The question is, "What does that tell us about the middle ground?" So this is an extreme case but what do we know about the middle case? Say your parent — You're not raised in a cage, you're not in a Romanian orphanage, but your parents just don't pick you up very much. They don't love you very much. There's no good evidence that that has any effect on a person. The problem is, and we're going to talk about this in much more detail in a couple of weeks, is it's true that parents who aren't affectionate have kids that aren't affectionate but it's not clear this is because of a genetic connection or an environmental connection. The one thing we do know is that in the middle ground, effects tend not to be dramatic. So when you get away from extreme cases, effects are hard to see and require careful experimental research to tease out. I think what it's safe to say for a lot — for everything but the severe conditions is we don't know what kind of effects there are. But if there are effects they are not big and dramatic ones. Okay.

Chapter 3. Evolutionary Explanations for Emotional Responses [00:18:16]

Animals' good feelings, animals' emotional attraction to their kin, is not a huge puzzle from an evolutionary point of view. Evolution is driven by forces that operate on the fact of how many genes get reproduced and replicated among your descendants. So, it makes sense that animals would be wired-up to care for their kids. It would make sense that kids who are wired-up to survive would develop attachments to their parents. What's more of a puzzle though is that animals, including humans, seem to have exquisitely complicated relationships with non-kin. In particular, animals are nice to non-kin. You are nice to people that you're not related to. There are a lot of examples of this. Animals groom one another. You go, you pick off the lice and the bugs off your friend; they pick it off you. They give warning cries.

So, warning cries — All sorts of animals give warning cries. You are — I don't know. You're a little animal and a big animal comes charging and you say, "Hey!" Oh. You may sort of cry and everybody runs away. And that's very risky for you but you do it anyway, often to protect people you aren't related to. Often animals share childcare. And from a cold-blooded, natural selection, survival-of-the-gene point of view, you would imagine that if you lend me your kid for the day I would eat him for the

protein and "it's not my genes and actually it gives more for my kids." That's not quite how it works though.

Animals share food. In fact, that animal, hugely ugly, the vampire bat, shares food. What happens is the vampire bat — vampire bats live in caves and they fly out. And what they do is often a bat will strike it big. She'll find a horse, for instance, bite the horse, pump in tons of blood and then fly back. And what it does is it doesn't keep it to itself. Rather, it goes around the whole cave and vomits blood into the mouth of all the other vampire bats so everybody benefits. Isn't that nice? [laughter] Now, what you're tempted to say is, "Well, that's really nice. Everybody benefits," but this raises a puzzle from the evolutionary point of view.

Remember, animals benefit more, and to this situation, animals benefit more by working together than by working alone. The benefits outweigh the costs. This is known as "reciprocal altruism" meaning my behavior to you, my good behavior to you, my altruism for you, is predicated on the idea of reciprocation, "I'll benefit from you." And you imagine how vampire bats, for instance, why this makes sense. This is — If you're a vampire bat, it's a better system when anybody strikes it big to feed you rather than for anybody who strike it big, use the blood and then spit out all the rest of it. But here's the problem. Here's why it's such a puzzle. The problem is the existence of cheaters. And in economics and sociology these are also known as "free-riders." And what a cheater or free-rider does is it takes the benefits without paying the cost. Imagine two genes. Imagine one builds a vampire bat that accepts blood from others and shares blood. The other gene accepts blood from others and doesn't share blood. In the long run, "B" will actually out-produce gene "A" because in fact, "B" will be healthy while other vampire bats get sick. And then so the offspring will do better.

An even sharper example is an example of warning cries. So, gophers give warning cries when there's a predator. It is extremely adaptive to give a warning cry. Sorry. It's extremely adaptive to respond to a warning cry. You hear a warning cry, you — "Oh, crap," you run away. It is not very adaptive to give a warning cry. A really good solution then is to listen to warning cries but not to give them. Suppose we had a system — It is very adaptive when people are going to the bar, when people buy drinks to accept the drinks. It is not so adaptive, from the standpoint of one's wallet, to buy people drinks. Here is a solution. Accept drinks but don't pay for drinks. And if everybody fell to that solution, the idea of buying a round would fade. So, there is the puzzle. Since cheating — Since a cheater, in the short run, can always out-win — does better than a non-cheater, how could this cooperation evolve? How could it be an evolutionarily stable strategy?

And the answer is "cheater detection." Reciprocal altruism can only evolve if animals are wired up to punish cheaters. Now, that requires a lot of mental apparatus. You have to recognize cheaters, you have to remember cheaters, and you have to be motivated to punish cheaters. And not every animal has this degree of complicated apparatus but actually we know that vampire bats do. So, in one clever study — So the theory says — the evolutionary theory says "yeah, I see what these vampire bats are doing," but you see — and this is a case where evolution makes a nice prediction that couldn't evolve unless bats are keeping track. If bats aren't keeping track, then the system could never exist because the cheaters would just take it over. They must be watching for cheaters.

So, the experiment which was done is you — a vampire bat strikes it big, it flies back, and then you keep it — as a scientist you keep it from giving blood to anybody else. Then what happens? Well, what

happens is when the other bats strike it big they starve the selfish bat, just as if we go to bars and everybody buys a round except for me. And this happens over and over and over again. Pretty soon you're going to buy a round but you're not going to give me one. And so, just as humans are keeping their eyes out for people who are taking the benefits without paying the costs, so are other animals. And it is argued that this sensitivity to cheating, this focus on reciprocation, plays a powerful role in the evolution of social behavior and the evolution of social emotions.

Chapter 4. Cooperative Behavior and the Prisoner's Dilemma [00:25:20]

And a classic illustration of this is The Prisoner's Dilemma. Now, many of you, I think, have seen The Prisoner's Dilemma in one course or another? It shows up — It is one of the main constructs in the social sciences. It shows up in cognitive science, psychology, economics, that you could — The teaching fellows are passing around something which you're not going to use right away. But for some of you this is the first time you're going to be exposed to The Prisoner's Dilemma so let me spell it out.

Here's the idea. You and a friend commit a crime. You rob a bank, for instance. For the sake of this example, you are prisoner two. You get caught. The police put you in a little room and they say, "We want to know everything that happened. In particular, we want you to rat out your friend." Now, here are the options, and one thing about this is nothing is hidden. The police officer could actually print out a copy of The Prisoner's Dilemma and put it right in front of you. And what he could say is, "Look." You're prisoner two. "You could cooperate — Well, you have two options. You could either cooperate with your friend, you could stay silent, or you could defect or you could squeal." But the police officer says, "Look. If you — Let me tell you something. If you cooperate with your friend and he squeals on you, you'll go to prison for life and he'll walk out. However, if you squeal on him and he cooperates, he keeps quiet, he'll go to prison for life and you'll walk out." So, what do you do? Now, on the nice side, what you can do is you could say, "No. I'm going to be quiet. I'm going to cooperate." Now, if you could trust your friend to cooperate, you're fine, you each get a little stint in prison, but of course your friend might defect. Your friend might squeal.

Here is the important structure of The Prisoner's Dilemma. No matter what you — what your friend chooses to do, you're better off squealing. So, suppose you're prisoner two. You believe your friend's going to cooperate with you. He's not going to be — he's not going to give the information out. Well, then your best thing to do is squeal on him. What if you believe he's going to squeal on you? Well, your best thing to do is squeal on him but if you could get your act together and you could coordinate this, you would both be quiet and get a fairly minor penalty. And you could see this — This is the standard origin of the prisoner's dilemma, why it's called "The Prisoner's Dilemma," but you could see this all over the place. So, here is the logic. The best case for you is to defect while the other person cooperates. The worst case is to cooperate while the other person defects.

Back to the police thing. The best case for you is to give up all the information; the other guy stays silent; you cut a deal; you walk home that day. The worst case is you're quiet, he cuts a deal, you go to prison for life, but overall the best is that each cooperate and overall the worst for both is if each defect. And the reason that makes this tragic is this. Regardless of what your opponent does, it pays to defect, but if both people defect both are worse off. I'll give a couple of other examples. No. [referring to a

slide] That's just to show that there's a cartoon corresponding to The Prisoner's Dilemma. It is that common.

Here's the idea. I am — I break up with my wife. We've been married for a while. We've decided we're not going to go through it together anymore and we break up. We're living in separate houses and we're starting to talk divorce. It occurs to me — Here's me. I put that out there. "Should I get a divorce lawyer?" I ask. Now, I know divorce lawyers are really expensive. And it's kind of difficult to get a divorce lawyer. But if I get a divorce lawyer — And so neither one of us get a divorce lawyer we'll just do okay. We'll get a mediator. We'll split the money down the middle. That'll be okay but I'm kind of tempted. If I get a divorce lawyer and she doesn't, my divorce lawyer will take everything she's got. I get everything, she loses everything. Maybe I should be nice. Hold it. What if she gets a divorce lawyer and I don't? Well, then I'll lose everything, she'll get everything. Well, we should both get a divorce lawyer then but we'd both do pretty badly.

Imagine we're two countries, country "A" and country "B." Should I do nuclear disarmament? That's pretty good. We'd do okay if both countries disarmed. We would live our lives; we'll raise taxes; we'll do whatever countries do. But wouldn't it be cool if I build up my weaponry and they don't? I'll invade, take everything they got. That's kind of tempting. Uh oh. Also, if I don't do anything and they do it, they'll invade my country, take everything. So, we both build up our weaponry and we both do pretty badly. Once you start thinking about things this way, there's no end to the sort of notions that could fall under The Prisoner's Dilemma.

A good example is a drug deal. Suppose I want to buy marijuana from you, or "reefer" as they call it on the street. [laughter] So, I have $1,000 and from you I'd like to buy a ton of reefer so — I'm rounding off. [laughter] So, you say, "Wonderful. Wonderful. Let's meet behind the gym, two in the morning on Friday, and we'll do the exchange. You bring $1,000, I bring the reefer." "Oh, cool. Okay. Good." And I think, "that's pretty good, a thousand bucks, I get the reefer, you get a thousand bucks. That's okay, that's the normal thing." But now something occurs to me. "Nobody's going to go to the cops if things go badly. So instead of doing — bringing the money, why don't I just bring a gun? You come with your reefer, I stick a gun in your face, take the reefer, go home."

Maybe I won't do that, but now I worry because you're thinking the same thing. So, you could show up with a gun, stick the gun in my face, take the thousand bucks, go home. I'll have no reefer. What will I smoke? [laughter] So, we both think this way. So, we both show up behind the gym, two in the morning, with guns. [laughter] Well, that's not as bad for either one of us if I had — I — you had a gun and I didn't have a gun. But still, we're both worse off than if we could cooperate and just do the damn trade. And so that's the structure of The Prisoner's Dilemma.

You can only appreciate The Prisoner's Dilemma by actually doing it. So, here's — here is a numerical equivalent to The Prisoner's Dilemma. Everybody should have a card in front of you, a file card. If you don't — If you didn't get a card, a piece of paper will do just as well. Please write on one side "cooperate" and on the other side "defect" and then please find a partner with whom to play one game. This is a one-shot game. One of you is player one. The player on the right-most side from my right could be player one. The other one is player two. Do you each have a partner? If you have three people, you could cluster together and do two and then two and just think. It is actually best if you've never

met or spoken to the person you're about to deal with. And the game is, when I say "go," simply show the person your choice.

To be clear, if you are player one and you cooperate and player two cooperates as well, you each get three dollars. If you are player one and you cooperate while player two defects, player two gets five dollars and you get bupkis and so on. On three, just show the card to your opponent, to your person you're playing with. One, two, three. [laughter] Okay. How many people in this room cooperated? How many cooperated? How many defected? [laughter] Okay. How many people are now five dollars richer? Okay. How many of you got nothing? [laughter] Okay. So, you're learning. You're learning that the person next to you is really an SOB. [laughter] Now, find the person next to you and you get to play again. And you get to play five games in a row. Play five games in a row and keep score. You just show it to each other, record the numbers, show it, show it, show it, show it. Go now. [laughter] Anybody here win twenty-five dollars? Yes, twenty-five? So you —

Student 1: He cooperated four times and I defected —

Professor Paul Bloom: That's twenty, twenty-one. [laughter] Okay. That's good. That's good. So, it really is a measure of honesty. [laughs] Anybody win twenty or more? Fifteen or more. Fourteen or less. Anybody do five or less? You're a good person. It's good. It's good. You played it with him?

Student 2: Yes.

Professor Paul Bloom: Bad person. [laughter] It's not really about good or bad. There was a great game once. It's a simple game, but it was a great game, a great, famous competition a long time ago, about 20 years ago, set up by the great computer scientist Robert Axelrod. And he put together a competition where people brought in computer programs to play this game, to play The Prisoner's Dilemma. And there were sixty-three competitors. And these computer programs were incredibly — Some of them were very simple, always be nice, always be — always cooperate, always defect. Some were elegant, prime number solutions and prototype responses, genetic algorithms crafted to figure out what the other person was doing and suss them out.

But the winner was developed by Anatol Rappaport. And Anatol Rappaport actually died about a month ago at quite an old age, a great scientist. What was interesting about this was he was the winner with his program but his program was also one of the simplest. It may well have been the simplest. It was called "Tit-for-Tat" and it worked very simple. It took four lines of basic code. The first time you meet a new program, cooperate. The first time you meet somebody, be nice. After that, do on each trial what the other program did on the previous trial. This beat sixty-two others.

And here is why. It had certain beautiful features. It starts friendly. Remember the best long-term solution is everybody's be — everybody's nice. It starts off nice but you can't — it's not a sucker. If you screw with it, it will defect back on the next turn. It is, however, forgiving. Do you want to get nice with it? Be nice. If you're nice, it'll be nice back at you later on. It's also transparent, nothing complicated about it, and that's actually important. It's not merely that it's not a sucker and forgiving. More to the point, it is — you could tell it's not a sucker. And you could tell it's forgiving. And this very powerful algorithm learned to cooperate even in the situation — and helped — learned to make it out the best even in a situation where there's a risk of cheating and betrayal.

Some psychologists have argued that our emotions correspond to the different permutations on The Prisoner's Dilemma. We like people who cooperate with us. This motivates us to be nice to them in the future much as the Tit-for-Tat algorithm says, "If you are nice to me now, I'll be nice to you back." We don't like being screwed with. We feel anger and distrust towards those who betray us. That motivates us to betray or avoid them in the future. And we feel bad when we betray somebody who cooperates with us. This motivates us to behave better in the future. You can break down the cells of The Prisoner's Dilemma in terms of emotions that they give rise to.

I did an experiment last night with my seven-year-old and my ten-year-old. I explained to them The Prisoner's Dilemma. I didn't give the divorce lawyer example but — [laughter] and we gave them a big thing of chocolate chips and — the good chocolate chips. We had the good chips and we had the matrix and we had them play. Now, what they did isn't so interesting, but what's interesting is they were furious at each other. One of them, the younger boy, was — kept being betrayed by the older boy including tricks like he'd say, "Okay. Let's both cooperate." "Yeah. Okay." Then he'd cooperate — "defect!" And [laughter] the response was anger, though not actually guilt on the part of the other boy [laughter] but rage. And we see these sort of things all the time in real life.

Chapter 5. The Ultimatum Game, Rationality and Irrationality [00:40:44]

You're familiar with The Prisoner's Dilemma but there's another game, which you might not be familiar with. It's called The Ultimatum Game. How many of you have encountered The Ultimatum Game? Okay, some of you. Very simple. Choose a partner. It's a very simple game. When economists study this they actually do this with real money. I do not have real money to let you do this too. One of you is "A," one of you is "B." The one on the right most from this side is "A." The other one is "B." Here is a very simple rule. I'd like "A" to turn to "B" and make an offer. "A" has ten dollars. You can give "B" any amount you choose from that ten dollars, from one dollar to ten dollars. "B" can do only one thing. "B" can accept it; if you accept it, you agree to take home the money and "A" keeps what ever's left — or reject it. If you reject it, you get nothing. Nobody gets anything.

Is everybody clear? So "A" is going to say, "I'll give you so and so dollars." "B" would say, "Okay," in which case "B" walks away with so and so dollars or — and "A" walks away with whatever rest or "B" could say, "Reject," in which case nobody gets anything. So, this game comes in two steps. The first thing: I would like "A" to turn to "B" and make your offer. Don't — "B" doesn't do anything yet. Make your offer. Your offer should be one word. People are explaining their offer. Make your offer. Okay. Stage two. Do not negotiate. [laughter] You're not — I see people waving their hands and it's complicated. It should be a number from one to ten, a positive integer. Now, "B" — I would like "B" to say one word and you can say it really loud on three. Accept or reject. One, two, three. [laughter]

Wow. How many people accepted? Anybody reject it? Good. Okay. How many people offered ten dollars? [laughter] How many people offered more than five dollars? Okay. How many people offered one dollar? Okay. When you offered one dollar did you accept? Anybody else offer one dollar? When you offered one dollar did your partner accept? Okay. How many people offered either four or five dollars? Okay. This is an interesting game because the person who offered — who accepted one dollar was being rational. One dollar is better than no dollars. So, the psychology of human rationality is such that, from a logical point of view, you should reason one dollar is better than nothing. A rational person

should accept one dollar. And because we're smart, a — you should offer one dollar but not many of you offered one dollar. Why? Because you knew people are not purely rational. People, even in a one-shot game, won't accept unfair distributions. They'll reject them just out of spite. And so, you need to offer more. And this has been studied from a neuro-economic point of view, which basically provides neuroanatomical evidence that people — if you offered them one dollar they get really pissed. [laughter] Nobody likes to be offered a dollar.

Now, there's a more general moral here, which is actually an interesting surprise of some relevance to everyday life. A rational person is easily exploited. A rational person's responses to provocations, to assaults will always be measured inappropriate. If you know I'm rational and you're in a sharing situation with me, you could say to me, "Hey. Here's a dollar. Hey, Mr. Rational, a dollar's better than nothing." "Well, okay," because I'm rational. Similarly, you could mess with me because you could harass me in all sorts of ways, take things that I own, as long as you reason that a rational person wouldn't start a fuss about this.

There is some advantage to being irrational, to having a temper. Because if you have a temper and you're known to be irrational, people are forced, by dint of your irrationality, to treat you better. Who am I going to take from? The person who's extremely reasonable or the person who has a hair-trigger temper? Well, I'm going to pick on a reasonable person because the unreasonable person might do unreasonable things. And this is faintly paradoxical, but often to be irrational, or at least to have a reputation for mild irrationality, gives you an edge.

Now, this isn't focus of provocation but this has also been presented in the theory of why people fall in love. Suppose you're choosing who to devote your life to, and it's a matter of huge trust. We're going to raise kids together. It's very important for you that I don't leave. And I am very rational so I say to you, "We should mate and have children because I find you the most attractive of everybody who was available that I've met so far. I'm very rational and so long as this continues to be the case we'll be together." Well, that's reasonable and rational but wouldn't you rather be with somebody who's head over heels in love? Head over heels in love is irrational but it's also, within certain parameters, endearing because the irrationality of the person means you could trust them more in the long run, just like the irrationality of somebody who has a temper means you don't mess with them as much.

The studies have been done more with regard to violence than with love. And in fact, the irrationality — the benefits of irrational violence have been translated in terms of the study of homicide and other crimes. Daly and Wilson describe the cause of murder. Most murder is not caused by reasonable provocation. Most murder is not rational in its response. Most murder is generated by insult, curse, petty infraction, but this is not crazy irrationality. It's adaptive irrationality. Daly and Wilson point out, "in chronically feuding and warring societies an essential manual — manly virtue is the capacity for violence. To turn the other cheek is not saintly, but stupid or contemptibly weak." If I show myself a rational person when picked on or harassed, I'll be known as somebody you could pick on and harass.

And in fact, it turns out even in the modern world — This is from a New York Times I just picked up a year ago today. And the point is that the violence is due to people disrespecting each other or giving a dirty look. And you might think "isn't that irrational?" But it's not irrational in circumstances where people live together in an environment where they have to deal with each other over and over again, and often where there's not much support by the police as indications they talked about here. What's

particularly interesting is this sort of importance of a reputation for violence differs from culture to culture. And I've been talking so far in this class — and in fact, so far in this course — about universals, about things that are built in, things that show up across humans and other animals. I want to turn now and end this lecture by talking a little bit about a cultural difference. And it's a psychologically interesting cultural difference with regard to the emotions. And it's built around the difference turning around what sociologists call "cultures of honor."

Chapter 6. Cultures of Honor [00:49:21]

A culture of honor has certain properties. You can't rely on the law. And it has resources that are easily taken. And sociologists have argued that when those conditions are met it becomes important to develop a reputation for violent retaliation. That becomes important. Examples of culture of honors include Scottish highlanders, Masai warriors, Bedouin tradesmen, and Western cowboys – all cases where there's resources such as cattle that are vulnerable and easily taken, but you can't count on calling 911 and having people come help you. But the culture of honor that's been studied the most by modern psychologists is the American South.

This was settled by herdsmen and traditionally has less centralized legal control. So, the sociologists say the American South has more of a culture of honor than the American North. But how do you know? What does that do? We're interested in this class in claims about psychology. So, it took Richard Nisbett and Dov Cohen to study cultures of honors and look at differences. And they found some interesting differences. Gun laws tend to be more permissive in the southern — in the American South than the American North. Corporal punishment and capital punishment tend to be more approved of. Attitudes towards the military are more positive. In questionnaire studies, people are more forgiving towards cultures of honor. Somebody insults my woman and I punch him in the face. This is considered less bad in the American South than the American North. There's a higher rate of violence but only in certain circumstances. The streets of the American South as a rule are not more dangerous than the American North. The difference is there's a higher rate of crimes that are crimes of honor such as, for instance, if somebody breaks in to my house, me shooting him. Or if somebody insults me, me killing him.

Now, this is sort of survey studies. So, Nisbett and Cohen did one of the more interesting psychological studies I have ever heard of. And they did this at — This — Sorry. This is Nisbett and Wilson. They did this with University of Michigan undergraduates. They did a subject pool thing like you're doing now, and on it your demographic information was listed. And what they did was they took white males who are not Hispanic and not Jewish. That was their sample. Culture of honor is a phenomena limited to males and they wanted to make it sort of a clean study so they wanted to focus — get a homogenous sample. So, not Hispanic, not Jewish. And they provoked them. And the provocation was genius.

What they did was they said — they brought people in to the psychology building, as you'd be brought in to Kirtland or SSS or Dunham [psychology buildings at Yale] and they said — they had somebody go in to the desk and they said, "Yeah. Go down the hall for the experiment." There was a hallway and then you walked through the hallway. And walking in the other direction at that moment a graduate student — a male graduate student would start to walk. And he's holding some files. And what he does is he bumps the person, looks at him and says, "Asshole" [laughter] and keep walking. Now, to be fair,

the graduate student survived bumping into hundreds of males, calling them assholes and then walking to — Fights did not break out, nobody was shot. But then they brought the men — now went in to a room and they were tested. And it turned out that there were differences in the stress response.

On average, males from the American South showed higher hormone response and stress response than males in the American North — increases in testosterone and cortisol. There's always differences in later behavior, the people — suggesting that they were made angry. They gave differences in fill-in-the-blank questions, for instance. I don't remember the examples but it's examples like "John went to the store and bought a 'blank'" and then the northerners would say "and buy an apple." And the southerners would say "an AK-47 [laughter] to kill that freaking graduate student." [laughter] Now, again, the American South — people in the American South were not overall more violent than the American North, but they were more sensitive to provocations of honor.

Now, when I gave this lecture a few years ago, a southern student contacted me afterwards and said that she felt that picking out the southern minority at Yale was in some regard offensive and that people say things at Yale about southerners — American southerners that they would never say about any other minority group. So, there's two points I want to make regarding this. One is, of course, these are average differences. Not every northerner and southerner would differ along these lines. But another one is I think the effect is real, but it's not entirely clear that it reflects poorly on the cultures of honor as opposed to the other cultures. So, Nisbett, for instance, is himself a southerner and he points out that he went to the North he was most astonished by how rude people are. And this is because the North — the American North is not particularly a culture of honor, and so there's less proper behavior towards other people because there's no fear of retaliation or response. Moreover, the culture of honor virtues like honor, loyalty, courage and self-reliance, are on the face of it not necessarily bad things.

In any case, this is an interesting example of how there's an evolutionary background but culture modifies and shifts it in different ways. More generally, I've suggested over the last couple of lectures that emotions like fear, the love you have towards your children, anger, gratitude are not aberrations or noise in the system. Rather, they're exquisitely complicated motivational systems that are crafted to deal with the natural and social environment. And we know this only from an analysis that starts from an evolutionary approach. So, to bring us back to D'Arcy Thompson, "everything is the way it is because it got that way." And your reading response for this week is that. [referring to a slide] And I'll wish you good luck on the exam on Wednesday. And I'll see you there.